

Linux Programming: Assignment-10: 23-11-2025

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ROLL NO- 04

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CLASS- CYBER SECURITY

SECTION- C

My USN is 21 so my commands are

Basic command: 21. nano – Simple text editor

Advanced command: 21. lsattr – Show file attributes on ext filesystems

1.What is the said commands (little history, invention, authors of the command)?

Ans:

1.Command: Nano

Nano is user friendly text editor like word and docx for unix(liunx)operating system

It is mainly used to view,create or esit any text files directly from the command line

It was created in 1999 by author “Chris Allegretta”

It was created to replace "Pico" a non open source pine email client

It was initially called TIP and renamed in 2000 to avoid name conflict

Original author is Chris Allegretta it is currently maintained by GNU project

License: GNU general public license

2.Command: lsattr

lsattr short end for list attribute used to display file attributes on ext2,ext3 and ext4 filesystem

These attributes define special behaviors for files

example making files immutable (cannot be deleted,modified or renamed)

lsattr command was introduced along with the ext2 filesystem which developed to improve performance and reliability over earlier linux file systems.

It was created by Remy Card it made its first appearance in 1990s as a part of the **e2fsprogs** package a set of tool designed to create,check and manage ext2/3/4 file system

Current maintainers :

e2fsprogs package and other contributors from linux foundation and open source community

license:

GNU general public license

2. Why the particular commands are useful for System administrator?

Ans

Command : nano

Nano has a easy user interface which help for beginners to make quick edits in file without learning any command

System admins sometimes have to edit

Or modify files like /etc/hosts, /etc/fstab, /etc/network/interfaces

But with nano command they can easily accesses and edit these files by command `sudo nano /etc/hosts`

As nano comes pre installed in linux systems it is easier to accesses

Nano is safe and reliable as it has undo option to prevent data loss

It supports multiple task at once

Command: `lsattr`

This inspects file level attributes on linux system on ext2, ext3, and ext4 filesystem

It helps in system security stability and intergrity by checking which files have special protection applied

Using `lsattr` admins can verfiy that these files are properly protected if it shows `-l` attribute the file cannot be deleted

Its also detects unauthorized modifications

Helps maintain system stability

File backup and recovery monitoring like `-d` attribute

3. How these said commands are working? Give proper example.

Ans

Command: `nano`

Nano command lie works as a simple text editor that load files directly to editable terminal that allows modification using intuitive shortcuts and saves changes to file

Example:

Syntax of nano command: `nano [filename]`

you can also add a modify behavior before filename

Creating a file:

Nano notes.txt

This will run the file if its already exists or creates the file if there is none

After opening you can type your content in the file

And to save changes we use intuitive shortcuts like

Ctrl + O to save chnages

Enter to confirm the filename and Ctrl +X to exit the file

We also use Ctrl + R to insert content from another file

Command: lsattr

Lsattr checks each file attributes stored by ext2/3/4 filesystem and prints them on terminal in a readable format

It displays special attributes set on files and directories these attributes define how files can be accessed,modified or deleted at the filesystem level

Syntax of lsattr:

Lsattr [option] [file or directories]

Example :

Lsattr myfile.txt

Output

```
----i-----e-- myfile.txt
```

Here I means that file is immutable(cannot be changed or deleted)

- Means other attributes are not set

e attribute shows the file uses extents

example 2:

lsattr -R /etc

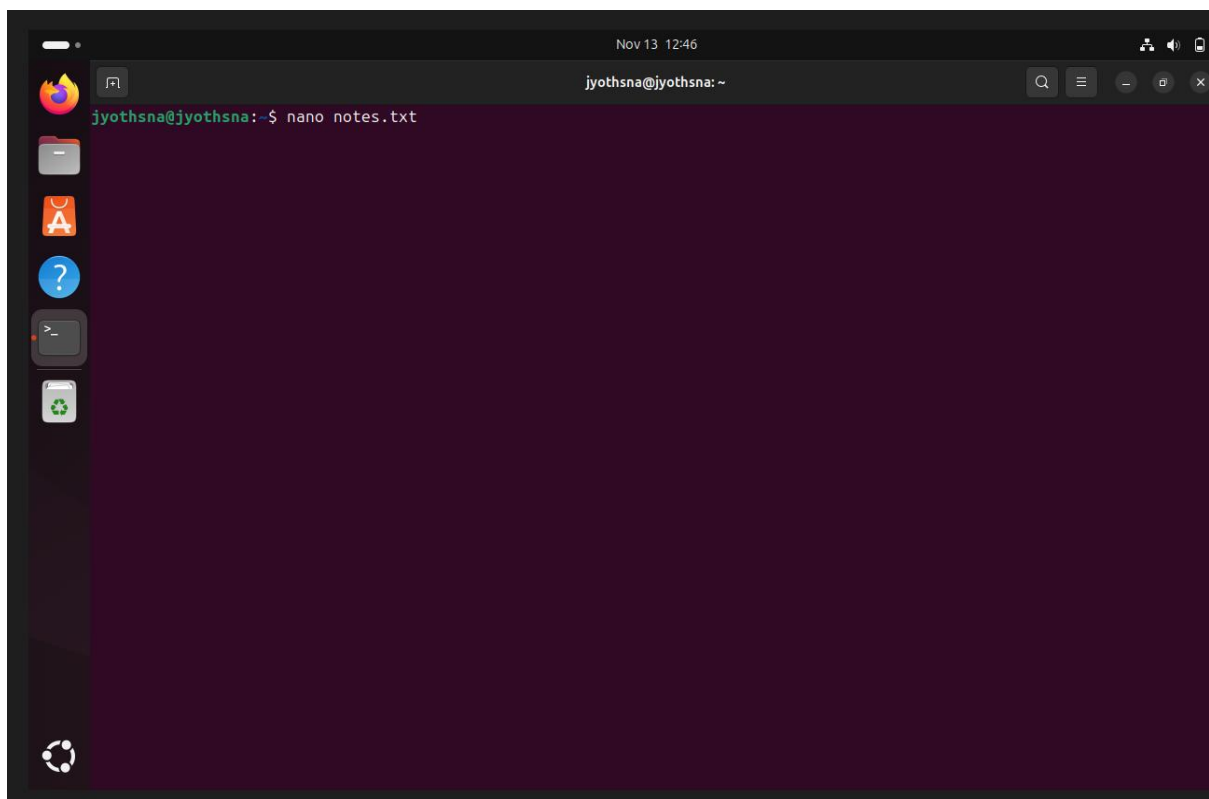
the option -R shows file attributes for all inside subdirectories

4. Few snapshots of the command execution with proper illustration.

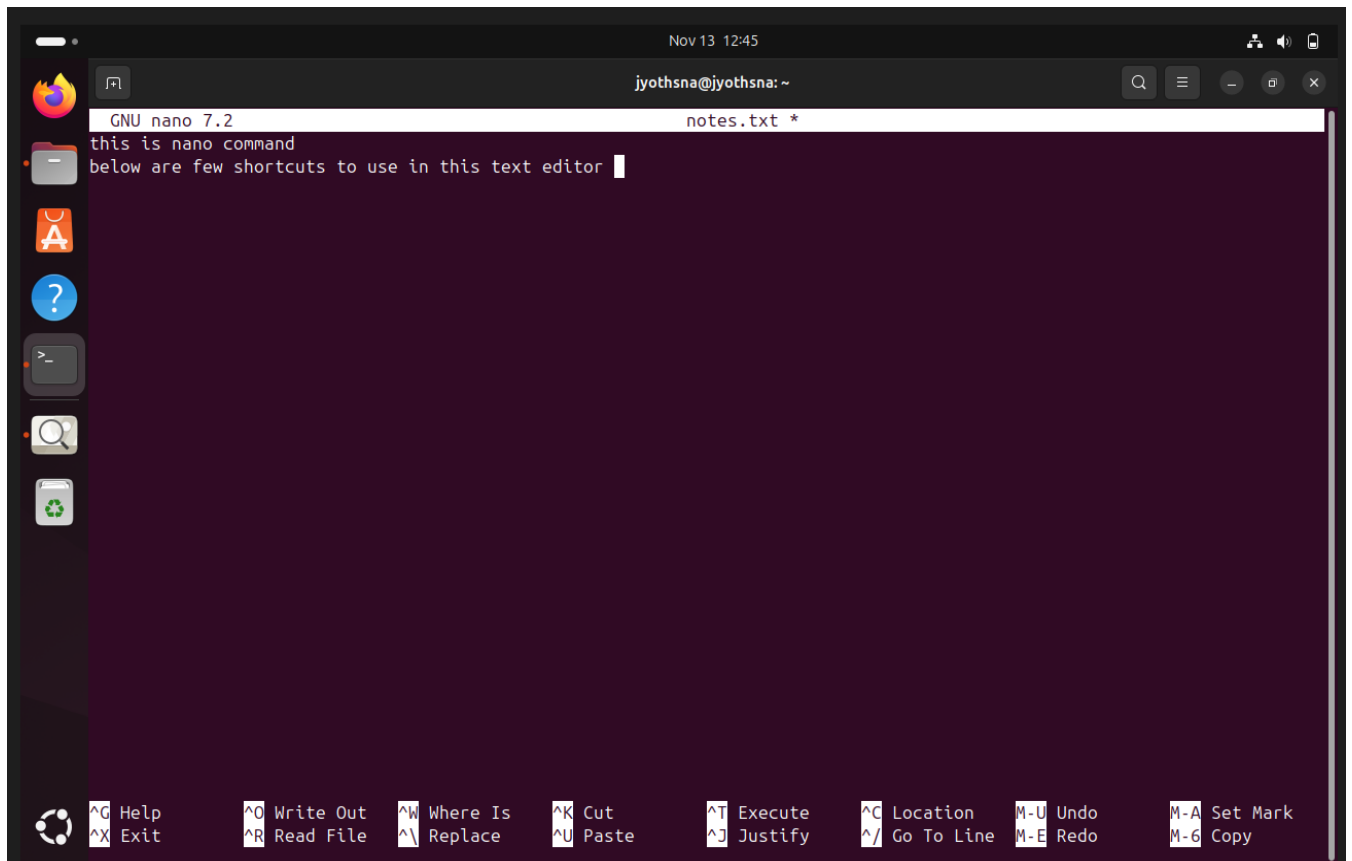
Ans

Command nano:

Command running on terminal :



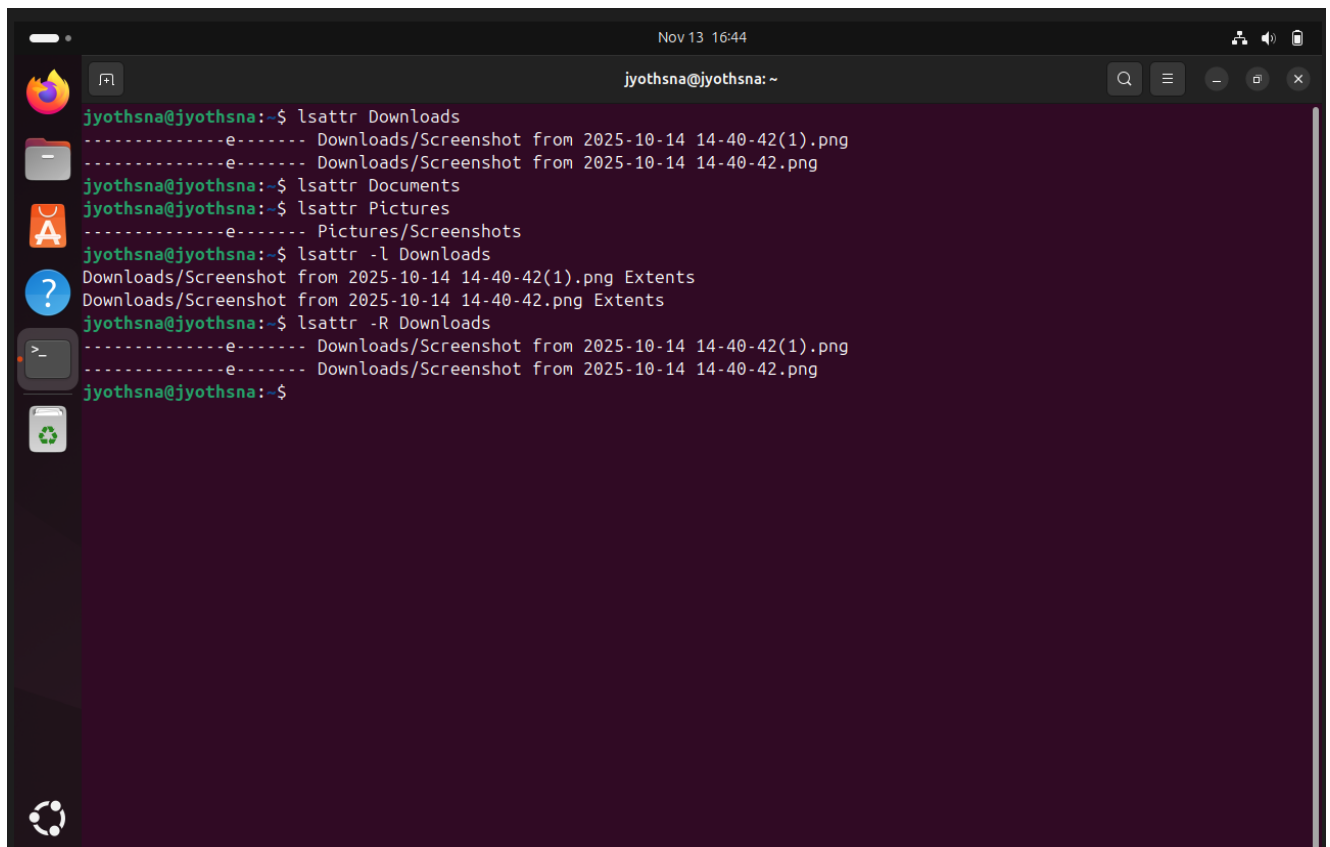
Created a text file named notes:



Shortcuts:

- Ctrl+G for help
- Ctrl+O to write in a file
- Ctrl+K to cut a line in file
- Ctrl+W where is to find
- Ctrl+T to execute a file
- Ctrl+C for a location of file
- Ctrl+X exit a file
- Ctrl+R to read a file
- Ctrl+\ for replace
- Ctrl+U to paste any line
- Ctrl+J justify
- Ctrl+/ go to line

Command : lsattr



```
jyothsna@jyothsna:~$ lsattr Downloads
-----e----- Downloads/Screenshot from 2025-10-14 14-40-42(1).png
-----e----- Downloads/Screenshot from 2025-10-14 14-40-42.png
jyothsna@jyothsna:~$ lsattr Documents
jyothsna@jyothsna:~$ lsattr Pictures
-----e----- Pictures/Screenshots
jyothsna@jyothsna:~$ lsattr -l Downloads
Downloads/Screenshot from 2025-10-14 14-40-42(1).png Extents
Downloads/Screenshot from 2025-10-14 14-40-42.png Extents
jyothsna@jyothsna:~$ lsattr -R Downloads
-----e----- Downloads/Screenshot from 2025-10-14 14-40-42(1).png
-----e----- Downloads/Screenshot from 2025-10-14 14-40-42.png
jyothsna@jyothsna:~$
```

5. At least elaborate five to six Option and Flags related to the said command. Give sources for detailed flags and options.

Ans:

The command nano provides many flags like -m,-c,-l,-b,-v and -e to make text editing more flexible, reliable, and efficient.

Flags or options:

-m: this enables the use of mouse inside the command as we cannot use mouse in nano text editor. By using this flag, we can use the cursor, select, and click menu items using mouse.

-c: this option tracks the cursor position and displays the current line number and column number of the cursor.

Useful when editing configuration or code files.

-i: this option indents new line to match the previous line's indentation or white space

It is useful to do shell scripts and code

-B: before saving any file nano creates a backup copy of the same file with a ~ symbol this helps in case something goes wrong in editing you will have a backup file to refer too

-v: this option only lets to view a file (view mode) making sure no accidental editing occur useful just to read any file

-E: converts each tab character into space keeps indentation consistent and useful

We can also combine multiple option together like

nano -c -m -i myfile.txt this opens nano with mouse support cursor position display and auto indentation

Source for detailed flags and option are:

<https://www.nano-editor.org/dist/latest/nano.html>

<https://www.gnu.org/software/nano/>

[Nano Text Editor in Linux - GeeksforGeeks](#)

Command: lsattr

Lsattr command includes several option that modify how files attribute are displayed these help system administrators get more detailed customized views of file attribute information

Options:

-a this option shows all the files including the hidden ones has a similar behavior too ls -a

Useful to check hidden configuration files

-d shows attributes of the directory itself not the file inside it it is useful for verifying if a directory is immutable or append only

-R is used for recursive this option attributes the file recursively including all the subdirectories

-l long name attributes displays long names instead of single letter code making its easier for beginners to understand what each attributes mean

-v shows the version information or prints the version of lsattr program and then exits

-p shows project hierarchy this displays project ID or hierarchy if the filesystem supports project quotas

We can also combine the flags together to create one flag for example: lsattr -aR /etc here the options used are -a and -R Here these option show all the hidden files and all the recursively subfolders as well

Source and detailed flags

<http://man7.org/linux/man-pages/man1/lsattr.1.html>

man command in the terminal can also be used

[chattr and lsattr commands in Linux with examples - GeeksforGeeks](#)

[lsattr Command in Linux \[5 Practical Examples\]](#)